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# CHM1001 2025 term 1

# Assignment 3

## 1. Multiple Choice Questions (Only one correct answer per question)

1. A gas sample occupies 3.00 L at 1.50 atm. What will the volume be at 1.00 atm, constant T and n?
   V P
   - A. 2.00 L
   - B. 3.00 L
   - C. 4.50 L
   - D. 1.50 L
2. The value of the gas constant R in $L \cdot atm \cdot mol^{-1} \cdot K^{-1}$ is:
   - A. 8.314
   - B. 0.08206
   - C. 62.36
   - D. 1.00
3. At STP, the molar volume of any ideal gas is:
   - A. 24.0 L
   - B. 1.00 L
   - C. 22.414 L
   - D. 0.08206 L
4. A gas has density 1.25 g/L at 1.00 atm and 273 K. Molar mass is:
   1.25 g
   $1 \cdot 1 = n \cdot 0.08206 \times 273$
   - A. 28.0 g/mol
   - B. 22.4 g/mol
   1.25 g → 0.044 mol
   - C. 31.2 g/mol

- D. 44.0 g/mol

5. Dalton's Law states that:
    - A. Total pressure is the sum of partial pressures
    - B. Mole fraction × partial pressure = total pressure
    - C. Gas volume proportional to temperature
    - D. Gas speed is proportional to molar mass
6. Which gas will have the highest root-mean-square speed at the same temperature?
    - A. $O_2$
    - B. $N_2$
    - C. $H_2$
    - D. $CO_2$
7. Graham's Law of Effusion states:
    - A. Rate ∝ molar mass
    - B. Rate $\propto \sqrt{(M_2/M_1)}$
    - C. Rate $\propto 1/\sqrt{M}$
    - D. Rate ∝ M
8. According to KMT, gas pressure results from:
    - A. Intermolecular attractions
    - B. Collisions of gas particles with container walls
    - C. Particle vibration
    - D. Gravity
9. Average kinetic energy of gas particles is proportional to:
    - A. Molar mass

- B. Volume
- C. Temperature in Kelvin
- D. Pressure

10. “Effusion” refers to:
    - A. Mixing of two gases without barrier
    - B. Escape of gas through a tiny hole without collisions
    - C. Gas flow through a tube
    - D. Gas dissolving in a liquid
11. Which IMF is present in all molecular substances, regardless of polarity?
    - A. Dipole–dipole
    - B. London dispersion forces
    - C. Hydrogen bonding
    - D. Ion–dipole interactions
12. Which has the strongest intermolecular forces?
    - A. $CH_4$
    - B. $NH_3$
    - C. $H_2O$
    - D. $CO_2$
13. Among $Br_2$, $I_2$, $Cl_2$, $F_2$, which has the highest boiling point?
    - A. $Br_2$
    - B. $I_2$
    - C. $Cl_2$

- D. $F_2$

14. Which of the following will experience hydrogen bonding between its molecules?

    - A. $CH_3CH_3$
    - B. $CH_3OH$
    - C. $CH_3Cl$
    - D. $H_2S$

15. Greater surface tension corresponds to:
    - A. Higher IMF strength
    - B. Lower IMF strength
    - C. Larger molar volume
    - D. Smaller molar mass
16. Which property decreases with increasing temperature?
    - A. Density of liquid
    - B. Vapor pressure
    - C. Surface tension
    - D. Volume of liquid
17. At the boiling point:
    - A. Vapor pressure = external pressure
    - B. IMF = 0
    - C. All molecules escape
    - D. Density = gas density
18. Using Clausius–Clapeyron, a plot of ln P vs. 1/T is:
    - A. Linear with slope = −ΔH_vap/R
    - B. Curved

- C. Linear with slope = ΔH_fus
- D. Linear with slope = −ΔH_fus/R

19. Sublimation is:
    - A. Solid → liquid
    - B. Liquid → gas
    - C. Solid → gas
    - D. Gas → liquid

20. In the phase diagram of $CO_2$, the triple point is:
    - A. Above 1 atm
    - B. Below 1 atm
    - C. At 1 atm
    - D. At critical point

21. The critical temperature is:
    - A. Above which gas cannot be liquefied regardless of pressure
    - B. Below which liquid cannot vaporize
    - C. Specific to perfect gases only
    - D. Temperature at triple point

22. Which has the highest viscosity?
    - A. $C_5H_{12}$
    - B. $CH_3OH$
    - C. $H_2O$
    - D. $C_8H_{18}$

23. Which phase change is exothermic?
    - A. Melting

- B. Freezing
- C. Vaporization
- D. Sublimation

24. Which statement about water is correct?
    - A. Ice is denser than liquid water
    - B. High heat of vaporization due to hydrogen bonding
    - C. Boiling point lowered under higher pressure
    - D. Expansion on freezing due to tightly packed structure
25. Which solid type typically has the lowest melting point?
    - A. Ionic solids
    - B. Molecular solids
    - C. Metallic solids
    - D. Covalent network solids
26. In a body-centered cubic (bcc) unit cell, the number of atoms per cell is:
    - A. 1
    - B. 2
    - C. 4
    - D. 6
27. In a face-centered cubic (fcc) unit cell, the number of atoms per cell is:
    - A. 1
    - B. 2
    - C. 4
    - D. 6
28. Packing efficiency of fcc is about:
    - A. 52%
    - B. 68%
    - C. 74%

- D. 100%

29. Which is an n-type semiconductor?
    - A. $\overset{4}{\text{Si}}$ doped with $\overset{3}{\text{B}}$
    - B. $\overset{4}{\text{Si}}$ doped with $\overset{5}{\text{P}}$
    - C. $\overset{4}{\text{Ge}}$ doped with $\overset{3}{\text{Ga}}$
    - D. $\overset{4}{\text{GaAs}}$ doped with $\overset{2}{\text{Zn}}$
30. Which is a p-type semiconductor?
    - A. $\overset{4}{\text{Si}}$ doped with $\overset{5}{\text{As}}$
    - B. $\overset{4}{\text{Ge}}$ doped with $\overset{5}{\text{Sb}}$
    - C. $\overset{4}{\text{Si}}$ doped with $\overset{3}{\text{Al}}$
    - D. $\overset{4}{\text{GaAs}}$ doped with $\overset{6}{\text{Te}}$
31. Metal conductivity is explained by:
    - A. Delocalized valence electrons in conduction band
    - B. Valence electrons fixed in bonds
    - C. Electron pairing
    - D. Ion movement
32. Diamond is an example of:
    - A. Metallic solid
    - B. Covalent-network solid
    - C. Ionic solid
    - D. Molecular solid
33. Sodium chloride is an example of:
    - A. Covalent-network solid
    - B. Metallic solid
    - C. Molecular solid
    - D. Ionic solid
34. In fcc, the relationship between edge length a and atomic radius r is:
    - A. a = 2r

- B. $a = 4r / \sqrt{3}$
- C. $a = 2\sqrt{2}\, r$
- D. $a = r / \sqrt{2}$

35. A covalent-network solid typically has:
    - A. High melting point, poor conductivity
    - B. High melting point, good conductivity
    - C. Low melting point, poor conductivity
    - D. Low density, high conductivity
36. Carbon nanotubes are:
    - A. Carbon atoms arranged in a flat sheet
    - B. Cylindrical structures of rolled graphene sheets
    - C. Hollow spheres of carbon
    - D. Crystals of graphite
37. An interstitial alloy occurs when:
    - A. Smaller atoms fill holes between larger metal atoms
    - B. Larger atoms replace smaller host atoms
    - C. Atoms are missing from lattice
    - D. Layers of atoms slide past each other
38. The band gap in semiconductors is:
    - A. Zero
    - B. Small, allowing limited conduction
    - C. Large, preventing conduction
    - D. Negative

## 2. Short-Answer/Calculation Questions

> 1. Calculate the molar mass of a gas with 5.00 g mass in 2.50 L at 27.0 °C and 0.850 atm.

From $PV = nRT$, we get

$$(0.850\ \text{atm})(2.50\ \text{L}) = n(0.082\ \text{L·atm/mol·K})(300\ \text{K})$$

$$n = 0.086\ \text{mol}$$

$$\rightarrow \mathcal{M} = \frac{5\ \text{g}}{0.086\ \text{mol}} = 57.88\ \text{g/mol}$$

2) $P_{He} = X_{He} \cdot P_{total}$

$= 0.4 \cdot 1.50 = 0.60$ atm

$P_{Ar} = P_{total} - P_{He}$

$= 1.50 - 0.60 = 0.90$ atm

2. A 2.50 mol mixture of He and Ar has a mole fraction of He ($X_{He}$) = 0.40. At a total pressure of 1.50 atm, find the partial pressures of He and Ar.
3. Using van der Waals constants for $CO_2$ (a = 3.59 $L^2 \cdot atm/mol^2$, b = 0.0427 L/mol), calculate the pressure of 1.00 mol of $CO_2$ at 320 K in a 5.00 L container.

   3) $\left(P + \frac{3.59 \cdot 1.00^2}{5.00^2}\right)(5.00 - 1.00 \cdot 0.0427) = 1.00 \cdot 0.082 \cdot 320$

   $\rightarrow P = 5.15$ atm
4. Compare the boiling points of $CH_4$, $CH_3CH_3$, and $CH_3OH$.

   Explain the trends.

   $CH_3OH > CH_3CH_3 > CH_4$

   Hydrogen bond ($CH_3OH$) stronger than dispersion force ($CH_3CH_3$), longer chain requires more energy to overcome the intermolecular force.
5. Explain why $NH_3$ has a higher boiling point than $PH_3$.

   ↳ $NH_3$ consists of hydrogen bonds which is stronger than dipole-dipole in $PH_3$. Thus $NH_3$ has a higher boiling point than $PH_3$.
6. Calculate the energy required to melt 100.0 g of ice at 0 °C

   (given ΔH_fus = 6.02 kJ/mol for $H_2O$). $Q = n\Delta H_{fus} = \frac{(100.0 g)}{(18 g/mol)}(6.02 kJ/mol) = 33.44$ kJ
7. Determine the edge length of bcc iron (given atomic radius r = 124 pm). BCC: $a = \frac{4r}{\sqrt{3}} = \frac{4(124)}{\sqrt{3}} = 286.37$ pm